Hyperemesis gravidarum

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Hyperemesis gravidarum describes nausea and vomiting in pregnancy severe enough to cause weight loss, dehydration, electrolyte imbalance, and nutritional deficiencies. The condition can render women so physically and mentally unwell that they are at increased risk of terminating a wanted pregnancy and experiencing suicidal ideation. Concerns regarding prescribing in pregnancy and inaccurate assumptions that the condition is self-limiting result in women being dismissed and having difficulty accessing appropriate care. Over the past decade, a wealth of literature has been published that gives new insights into the causes of hyperemesis gravidarum, the safety of antiemetic therapy, and short-term and long-term consequences for women with the condition and their children. This Review summarises the findings of this literature with the aim of informing decisions about the care of these women and future research priorities.

Introduction

Nausea and vomiting of pregnancy (NVP) is common, affecting 70–80% of the pregnant population.[1,2] Hyperemesis gravidarum describes a severe form of NVP that affects 0·3–3·6% of pregnant women.[1] In 2021, the Windsor definition (an international consensus definition) for hyperemesis gravidarum was developed, consisting of: symptom onset before 16 weeks gestation; nausea and vomiting, one of which must be severe; inability to eat or drink normally (or both); and greatly restricted daily activities.[3] Signs of dehydration were deemed contributory to the definition. The lack of a consensus definition before 2021 hampers comparisons with earlier studies, which used varying definitions for the condition. Hyperemesis gravidarum represents the leading cause of hospital admission in the first trimester of pregnancy and the second leading cause (after preterm birth) of antenatal hospital admissions overall, and is associated with substantial costs for health services.[4–6] The condition is understudied, with few studies to inform its management. This paucity of literature, combined with the condition often being dismissed by health-care professionals, results in adverse outcomes for women.[7,8] In this Review, we summarise the literature with the aim of informing best practice.

Epidemiology

A personal and family history of hyperemesis gravidarum are notable risk factors for the condition. A systematic review reported rates of hyperemesis gravidarum recurrence in subsequent pregnancies to be between 15% and 81%.[9] In a large Californian study, longitudinally linked medical records were used to examine the risk of hyperemesis gravidarum recurrence in second and third successive pregnancies.[10] The risk of hyperemesis gravidarum in the second pregnancy was 8·2-fold higher in women who had a history of the condition compared with those who did not; recurrence risk was 10·8-fold higher in women who had previous severe disease (defined as metabolic disturbance requiring inpatient management). Recurrence rates were increased in all ethnicities but were highest (34·4%) in women with African American ancestry. Recurrence risk was unaffected by the interpregnancy interval. Having two consecutive pregnancies with hyperemesis gravidarum in any ethnicity was associated with a 30-fold increased risk of a third pregnancy with the condition. 50% of these third pregnancies were complicated by hyperemesis gravidarum.

In a UK population-based cohort including 8·3 million pregnancies, risk factors for hyperemesis gravidarum included younger maternal age, lower socioeconomic class, being of Asian or Black ethnicity, carrying a female fetus, multiple pregnancy, and history of type 1 diabetes.[11] Additional risk factors include a personal history of motion sickness or seasickness, and pregnancy after assisted reproductive techniques.[12,13]

Search strategy and selection criteria

We searched PubMed, MEDLINE, Embase, and the Cochrane Library for papers published between Jan 1, 2000, and Feb 1, 2025, with the search terms "hyperemesis", "nausea and vomiting", "pregnan*", "morning sickness", and "hyperemesis gravidarum". We also searched the reference lists of papers of interest. Although we focused on publications between 2015 and 2025, we also referenced important older publications. The search strategy identified 6383 manuscripts, of which 91 were included in the review.

Aetiology

Hyperemesis gravidarum has a multifactorial aetiology that includes genetic, endocrine, and gastrointestinal factors, but not psychological factors.

Genetics

Initial evidence for a genetic component of the aetiology of hyperemesis gravidarum included a questionnaire study of affected women and controls that found significantly increased rates of the condition in mothers and sisters of affected women,[14] twin studies showing increased rates of severe NVP in monozygotic and dizygotic twins,[15,16] and population registry studies of the likelihood of children of mothers with hyperemesis gravidarum also having the condition.[17] Initial genome-wide association studies of women with hyperemesis

gravidarum or NVP (or both) identified two loci that were significantly associated with severe forms of the condition—*GDF15* and *IGFBP7*—and also showed significant associations with genes for the receptors of GDF15 and GFRAL, and the progesterone receptor gene, *PGR*.[18] These associations have been confirmed in a multi-ancestry, genome-wide association study of severe NVP that included more than 10 000 cases and 461 461 controls of European, Asian, African, and Latino ancestry.[19] An exon sequencing study of 926 women with hyperemesis gravidarum and 660 controls also identified a coding variant (C211G) within *GDF15* in ten cases that was predicted to be pathogenic.[20]

GDF15 is a hormone produced as a cellular stress response in non-pregnant individuals and is also one of the most abundant proteins secreted from trophoblast organoids.[21] Concentrations of GDF15 rise rapidly in early pregnancy and cause taste aversion, nausea, and vomiting. GDF15 was shown to be present in higher concentrations in serum samples from women admitted to hospital with hyperemesis gravidarum compared with controls with no NVP.[22] Subsequent studies have shown that *GDF15* risk variants for hyperemesis gravidarum are associated with reduced serum GDF15 in non-pregnant individuals,[23] and the authors hypothesise that mothers carrying *GDF15* risk variants have reduced circulating GDF15 concentrations before conception, and are therefore likely to be more sensitive to placentally secreted GDF15 in pregnancy. Supporting data indicate that future studies on blocking people's response to GDF15, blocking the secretion of GDF15, or desensitising people with a genetic predisposition for low GDF15 before they become pregnant, could be effective treatment strategies for hyperemesis gravidarum in the future.

Endocrine factors

A temporal relationship exists between human chorionic gonadotropin (HCG) concentrations in early pregnancy and the timing of NVP or hyperemesis gravidarum symptoms. This relationship historically led to the hypothesis that HCG has a central role in the aetiology of NVP and hyperemesis gravidarum; however, evidence supporting this association has not been definitive. In a systematic review and meta-analysis of 81 articles exploring diagnostic biomarkers for the presence or severity of NVP or hyperemesis gravidarum, 35 studies focused on the role of HCG, of which 18 found a significant association between raised HCG and occurrence of NVP or hyperemesis gravidarum, and 13 did not.[24] However, no clear biological plausibility for causation was apparent. Similarly, many studies have focused on thyroid hormones, and some have investigated leptin, oestradiol, and progesterone concentrations.[25–28] The results of these studies were not consistent, suggesting that these endocrine factors do not have a major role in the aetiology of hyperemesis gravidarum. The genetic association between the condition and the progesterone receptor gene suggests a link between hyperemesis gravidarum and progesterone signalling. Furthermore, authors have observed that women taking supplemental progesterone in the context of pregnancies conceived using assisted reproductive technologies often experience severe hyperemesis gravidarum that improves once supplemental progesterone is discontinued.

Gastrointestinal causes

The role of *Helicobacter pylori*, a Gram-negative bacterium, also features as one of the proposed mechanisms for NVP and hyperemesis gravidarum. The aforementioned systematic review and meta-analysis identified 32 articles that mentioned *H pylori*, of which 13 found no association between *H pylori* and the risk of hyperemesis gravidarum, and 19 suggested the bacterium could have a role in the condition.[24] The presence of *H pylori* is more probable in Black women than in White women, which could explain the increased rate of *H pylori* in women with hyperemesis gravidarum, but a direct role in causation of the condition is unlikely. As noted earlier, women with type 1 diabetes have an increased risk of hyperemesis gravidarum,[11] which likely relates to gastroparesis (a long-term condition of diabetes mellitus).

Clinical features

Symptoms of nausea and vomiting typically start between the fourth and seventh week of gestation, peak around the ninth week, and, in 90% of women, resolve by the 20th week.[2] In a prospective study of 363 women, 292 documented the time of day in which they experienced symptoms of nausea and vomiting; 91% (12 065/13 302) episodes of nausea and 47% (1901/4071) episodes of vomiting were after 1200 h, showing that symptoms can occur throughout the day and are not confined to the morning.[2]

In a US study of 819 women with hyperemesis gravidarum[30] (defined as weight loss and debility secondary to NVP, typically requiring pharmacological management), 50·3% (412/819) reported that their worst symptoms occurred in the first 12 weeks of pregnancy, but 22·0% (180/819) reported symptoms persisted throughout gestation. Weight loss of >5–10% of pre-pregnancy weight was reported in 27% (221/819) of women, >10–15% in 31% (254/819), and >15% in 26% (214/819). Women with extreme weight loss (>15% of pre-pregnancy weight) were more likely to have symptoms of hyperemesis gravidarum throughout gestation.

Ptyalism (excess saliva) is the second most reported symptom of hyperemesis gravidarum after nausea and vomiting, affecting three-quarters of women with the condition.[30] The symptom is distressing, with women reporting excess saliva volumes of 2 L per day, but has not been associated with specific risks to the mother or fetus.[31] In a study of 278 women with NVP, ptyalism was more common in Black women (73·1%; 19/26) than in

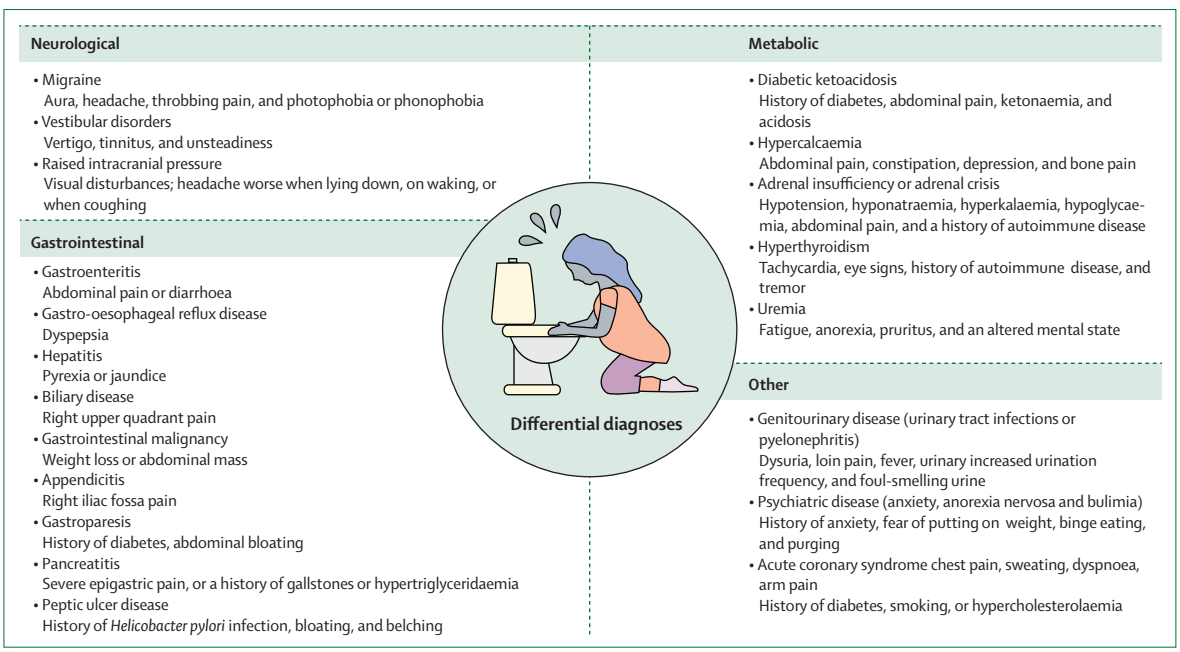

Figure 1: **Differential diagnosis of nausea and vomiting in pregnancy**

	Rationale
Initial investigation	
Full blood count	A raised white cell count could indicate infection and possibly indicate an alternative cause of symptoms; a raised haemoglobin or haematocrit count could indicate dehydration
Urea and electrolytes	To assess for hypokalaemia or hyperkalaemia, hyponatraemia, or acute kidney injury
Blood glucose	A raised blood glucose could indicate diabetes as a cause of symptoms
Urine dipstick or midstream specimen of urine	Nitrites could indicate a urinary tract infection; ketones are not a marker of dehydration and do not have a role in the investigation or management of hyperemesis gravidarum
Pelvic ultrasonography	To assess whether the pregnancy is viable and intrauterine; to screen for a multiple pregnancy; and to rule out trophoblastic disease
Investigations for selected or refractory cases	
Thyroid function tests	To exclude overt thyroid disease as a cause of symptoms
Liver function tests	To exclude liver pathology as a cause of symptoms
Bone profile	To rule out hypercalcaemia as a cause of symptoms; and to check for hypocalcaemia or hypophosphataemia
Amylase	To exclude pancreatitis as a cause of symptoms
Venous blood gas	To exclude metabolic disturbance

Table: **Recommended investigations for women with hyperemesis gravidarum**

White (19·4%; 46/237) or Hispanic (31·3%; 5/16) women.[32]

Gastro-oesophageal reflux disease (GORD) is reported in 64% of women with hyperemesis gravidarum.[30] Complications of GORD can include erosive oesophagitis, bleeding, and strictures, but these are rare. Health-care providers should be vigilant and screen for symptoms that could indicate a more serious cause.

Clinical assessment and investigations

Clinical assessment should include a careful history, symptom assessment, and consideration of other possible differential diagnoses (described in figure 1). Hyperemesis gravidarum is a diagnosis of exclusion. Examination should include an assessment of hydration and nutrition to help rule out other causes of nausea and vomiting. Investigations (outlined in the table) should also be used to exclude alternative diagnoses. The severity of symptoms can be quantified by use of the pregnancy-unique quantification of emesis (PUQE) or hyperemesis-level prediction (HELP) score. The PUQE score assesses symptom profile over 24 h and is a valuable tool for assessing severity, response to treatment, and nutritional intake.[33,34] The HELP score is a validated 12-question tool for screening and scoring levels of NVP that might be more accurate for patients at the severe end of the clinical spectrum.[35]

Systemic features and complications

Untreated hyperemesis gravidarum is associated with several systemic features and complications (summarised in figure 2). These complications can be serious; in a systematic review, the most frequent life-threatening complications included Wernicke's encephalopathy, electrolyte imbalance resulting in cardiac arrhythmia, and vitamin K deficiency causing coagulopathy.[36]

Biochemical thyrotoxicosis

A recent systematic review and meta-analysis[37] confirmed an association between hyperemesis gravidarum and transient biochemical hyperthyroidism, which can be

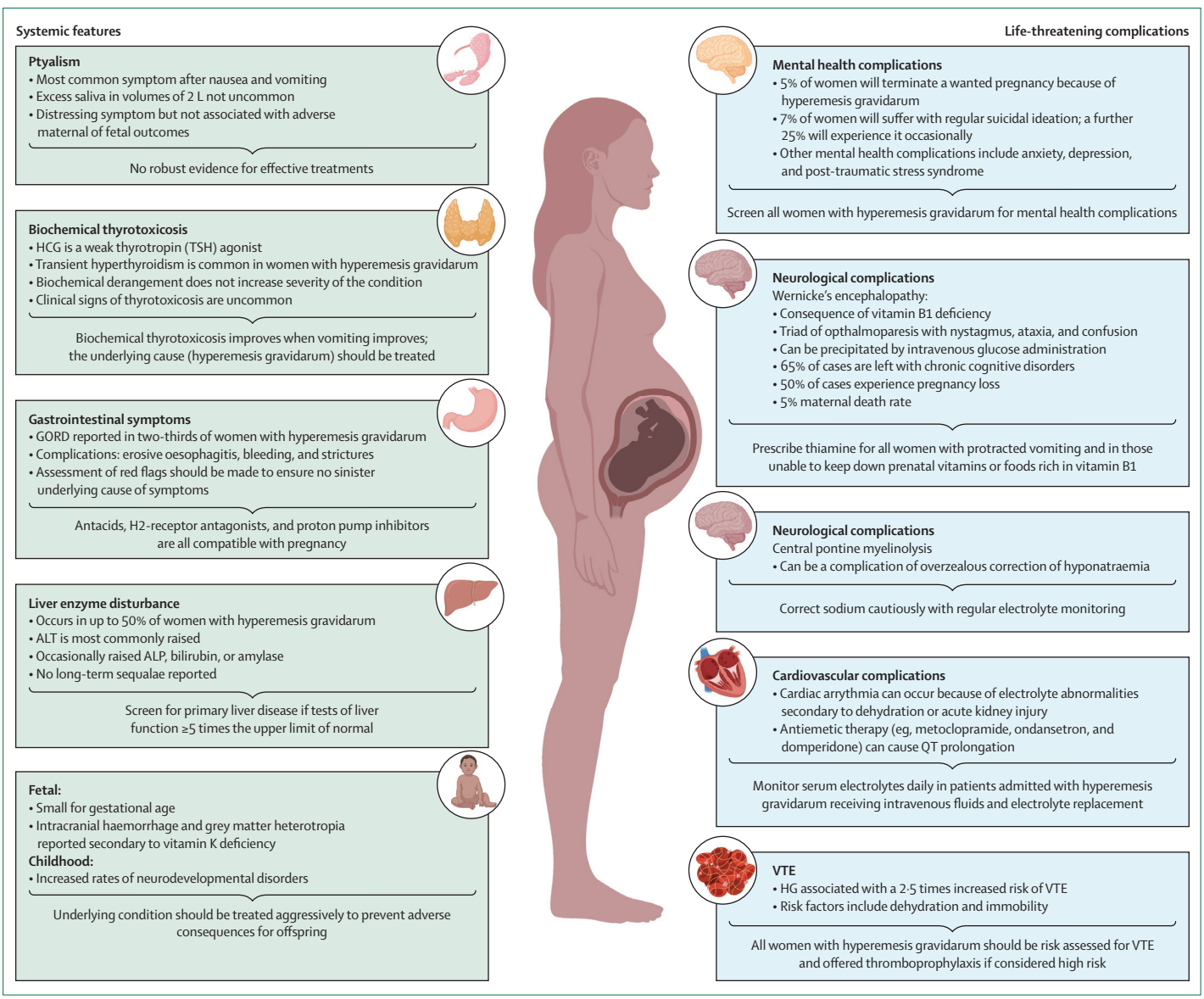

Figure 2: **Clinical features and complications of hyperemesis gravidarum**
ALP=alkaline phosphatase. ALT=alanine aminotransferase. GORD=gastro-oesophageal reflux disease. HCG=human chorionic gonadotrophin. TSH=thyroid-stimulating hormone. VTE=venous thromboembolism. Figure created with BioRender.com.

anticipated in up to 66% of women with the condition.[38] HCG and thyroid-stimulating hormone (TSH) share an α subunit; as such, HCG can bind the TSH receptor to stimulate thyroxine production. The degree of biochemical derangement does not predict the severity or clinical course of the condition.[39] Although some symptoms can occur in both hyperemesis gravidarum and thyrotoxicosis (eg, tachycardia and weight loss), very few women will have other symptoms of hyperthyroidism,[25] such as diarrhoea, tremor, or heat intolerance. The presence of these symptoms or other features of Graves' disease should prompt further investigation with TSH receptor antibodies.

Liver test disturbance

Serum liver test disturbances occur in up to 50% of women with hyperemesis gravidarum.[40] Proposed mechanisms include starvation injury and dehydration. Mild aminotransferase elevation of up to 200 IU/L is the most common disturbance, but increased alkaline phosphatase and amylase, and hyperbilirubinaemia, have also been reported.[41] No long-term sequelae have been recorded. In the absence of concerning features, the European Association for the Study of the Liver's clinical practice guidelines on the management of liver diseases in pregnancy recommends screening for primary liver disease only in those with marked

increases in liver enzymes (≥5 times the upper limit of normal).[42]

Complications

Nutritional deficiencies

As described earlier, vitamin B1 deficiency can result in Wernicke's encephalopathy. Vitamin K deficiency has also been described in women with hyperemesis gravidarum and, in some cases, has resulted in coagulopathy-related haemorrhage.[43] Other nutrients have not been thoroughly investigated but the existing literature indicates that the other deficiencies probably occur.[44]

Cardiovascular complications

Cardiac arrythmia has been reported as a complication of hyperemesis gravidarum in women with electrolyte imbalances.[36] Ventricular arrythmias have been described as QT prolongation resulting in torsade de pointes, which can occur as a result of hypokalaemia, hypomagnesaemia, and hypocalcaemia; antiemetic therapies such as metoclopramide, ondansetron, and domperidone given without adequate electrolyte replacement can also be contributory.

Venous thromboembolism

Because hyperemesis gravidarum is associated with dehydration and immobility, the risk of venous thromboembolism is increased above that associated with pregnancy. One study reported an odds ratio [OR] of 2·5 (95% CI 2·0–3·2) for the risk of developing a venous thromboembolism.[45] In the UK's 2020–22 confidential inquiry into maternal deaths, 16 women died from venous thromboembolisms in the first trimester of pregnancy, of whom six (37·5%) had hyperemesis gravidarum.[46]

Neurological complications

Neurological complications of hyperemesis gravidarum are rare but potentially fatal. Central pontine myelinolysis occurs most frequently after rapid correction of hyponatraemia and has been described in some women with hyperemesis gravidarum.[36] Wernicke's encephalopathy occurs because of vitamin B1 deficiency and is characterised by ophthalmoparesis with nystagmus, ataxia, and confusion. In a systematic review including 177 pregnant women,[47] treatment with subtherapeutic thiamine doses was common (64% of women; 113/177) and Wernicke's encephalopathy was precipitated by intravenous glucose administration in 14% (25/177). Wernicke's encephalopathy resulted in chronic cognitive disorders in 65% (115/177) of cases, pregnancy loss in 50% (89/177), and maternal death in 5% (9/177).

Mental health complications

Hyperemesis gravidarum has been shown to result in severe psychological distress, with women experiencing increased rates of anxiety[48] and depression[49] and lower quality of life.[50] In a study of 5071 women with hyperemesis gravidarum, 26% (1264/4964) reported occasional and 7% (331/4964) reported regular suicidal ideation as a result of their symptoms.[7] Suicidal ideation was associated with severity of sickness, poor functional status as a result of symptoms, and a perception of poor quality of care. Post-traumatic stress syndrome (PTSS) is common after a pregnancy with hyperemesis gravidarum; in a study of 377 women with the condition, 18% (68/377) met the full criteria for PTSS.[51] Although hyperemesis gravidarum is associated with mental health complications, contrary to historic belief, psychogenic causes do not feature in the aetiology of the disease.

Gestational disorders

Studies have described increased rates of pre-eclampsia and gestational diabetes in women with hyperemesis gravidarum.[52,53]

Outcomes for the offspring

The largest threat to the healthy survival of offspring in pregnancies affected by hyperemesis gravidarum is the increased chance of pregnancy termination: 5–15% of women with the condition terminate their wanted pregnancy.[7,54] The magnitude of this risk places hyperemesis gravidarum among the pregnancy conditions most lethal to offspring survival (despite being less well known) in contrast to pre-eclampsia, which leads to stillbirth in 0·5% of pregnancies.[55]

NVP, which affects the majority of pregnancies, is often considered a welcome sign of a healthy pregnancy and has been reported to be associated with a reduced risk of miscarriage.[56] Recent evidence suggests the opposite could apply to hyperemesis gravidarum,[56] with specific conditions (eg, Wernicke's encephalopathy) being associated with an increased risk of fetal loss[47] and vitamin K deficiency reportedly resulting in fetal intracranial haemorrhage and grey matter heterotropia.[43] Moreover, hyperemesis gravidarum is associated with increased rates of preterm birth (OR 2·81, 95% CI 1·69–4·67) and low birthweight (OR 1·43, 1·02–1·99 for birthweight <1500 g),[57] and could also be associated with an increase in cardiovascular anomalies.[22] These effects are probably at least partly due to insufficient maternal pregnancy weight gain;[58] however, the extent to which poor maternal nutrition or weight gain affects perinatal outcomes is unclear.

Management

The first step in managing hyperemesis gravidarum should be to listen to the woman's concerns and validate her symptoms. Lifestyle measures such as taking rest, dietary advice (ie, eat little and often), and encouraging maintenance of oral hydration (if possible) should be implemented. If lifestyle measures do not control symptoms, care in the community with antiemetics

should be offered.[59] Women might also benefit from rest and time off work, and consideration should be made as to the financial consequences of being unable to work for protracted periods.

In addition to antiemetic therapy, women with severe NVP or hyperemesis gravidarum will usually require intravenous fluid and electrolyte replacement.[60] If virtual wards (ie, hospital care provided at home), acute care at home, or outpatient services are available to women with hyperemesis gravidarum, these should be used to prevent the need for admission to secondary care centres and improve patient satisfaction.[61,62] Inpatient care with parenteral antiemetics should be considered if any of the following exist:[63] continuous NVP and inability to keep down antiemetics in health-care settings where rectal or sublingual routes are unavailable; continuous NVP associated with clinical dehydration and weight loss (>5% of bodyweight) despite oral, rectal, or sublingual antiemetics; confirmed or suspected comorbidity (eg, urinary tract infection or an inability to tolerate oral antibiotics); and comorbidities such as epilepsy, diabetes, HIV, hypoadrenalism, or psychiatric disease, where inability to tolerate essential oral medication could present further complications.

The management of hyperemesis gravidarum is summarised in figure 3. This figure is adapted from the Royal College of Obstetricians and Gynaecologists' green-top guideline on the management of NVP and hyperemesis gravidarum, which is based on a combination of evidence-based recommendations and expert opinion.[63] Best practice dictates that women should only be discharged from hospital once they are able to tolerate oral antiemetic therapy, nutrition, and hydration. However, this aim might not be achievable for women with severe disease, and some might prefer to be discharged home once adequately hydrated, accepting that they may require readmission or access to an ambulatory unit to receive ongoing treatment. All care should be culturally sensitive, particularly for those with multimorbidity or social deprivation, minority ethnic women, and for women whose first language is not English.

Historically, clinicians working in both primary and secondary care have been reluctant to prescribe early, adequate, and multiple antiemetics because of unfounded concerns regarding teratogenic risk.[64,65] The Royal College of Obstetricians and Gynaecologists' green-top guideline gives suggested first-line, second-line, and third-line antiemetics (figure 3). Given that women have substantial concerns regarding the safety of medications in pregnancy, reassurance that antiemetics can be safely used to control symptoms and prevent adverse effects of hyperemesis gravidarum is recommended.[8]

The recommendations for choice of antiemetics are based largely on safety data rather than efficacy data, as most randomised controlled trials (RCTs) comparing different antiemetics have notable limitations, including being single-centre studies and having small sample sizes. Comparisons are limited by the quality and heterogeneity of evidence. A systematic review found that the quality of many studies focused on antiemetic treatment for hyperemesis gravidarum was downgraded due to clinical heterogeneity, imprecision, sparseness of data, or a combination of these factors.[65] The initial assessment and subsequent reporting of nausea, vomiting, and other relevant outcomes also varied considerably.[65] Furthermore, very few double-blind RCTs have been designed to assess specific antiemetics versus placebo in women with hyperemesis gravidarum.

Most safety and efficacy data exist for H1-antihistamines, phenothiazines, and pyridoxine–doxylamine combinations, hence their inclusion in guideline-recommended first-line therapies. A Cochrane review and a systematic review exploring the safety and efficacy of these antiemetics reported no safety signals, no increased risk of teratogenesis, or other adverse pregnancy outcomes.[66,67] Dopamine antagonists, including metoclopramide and domperidone, were also found to have no adverse effects.[66,67] These dopamine-2 receptor antagonists are recommended as second-line treatment options because of their association with an increased risk of neurological side-effects, such as extrapyramidal disorders and tardive dyskinesia, when used outside pregnancy.

Ondansetron is an extremely effective antiemetic when used outside pregnancy. Data for its efficacy in hyperemesis gravidarum come from small, single-centre RCTs and suggest that it is more effective than H1-antihistamines and metoclopramide for managing the condition.[68,69] However, concerns have been raised regarding a possible association with isolated cleft palate and cardiac defects,[70] but available data are conflicting. If a causal association exists between ondansetron and cleft palate in an infant, the absolute risk increases to approximately 0·14% compared with a background risk of approximately 0·11%. The studies describing a small increased risk of cardiac septal defects are also conflicting.[71] Furthermore, not all studies have used disease-matched controls, and malnutrition and micronutrient deficiency could possibly explain the association with malformations in some studies. Overall, systematic reviews and meta-analyses have concluded that no significantly increased risk exists for major malformations, heart defects, orofacial clefts, genitourinary malformations, or hypospadias with first trimester exposure to ondansetron.[72,73] The authors of these studies believe that the benefit–harm ratio of ondansetron (based on extrapolation of its efficacy outside the context of hyperemesis gravidarum) is in favour of its use among women with the condition who do not respond to first-line antiemetics. When used, ondansetron should be co-prescribed with a laxative, as constipation is a common side-effect of the medication.

Third-line therapy with corticosteroids, which are both safe and effective,[73] should be reserved for women whose

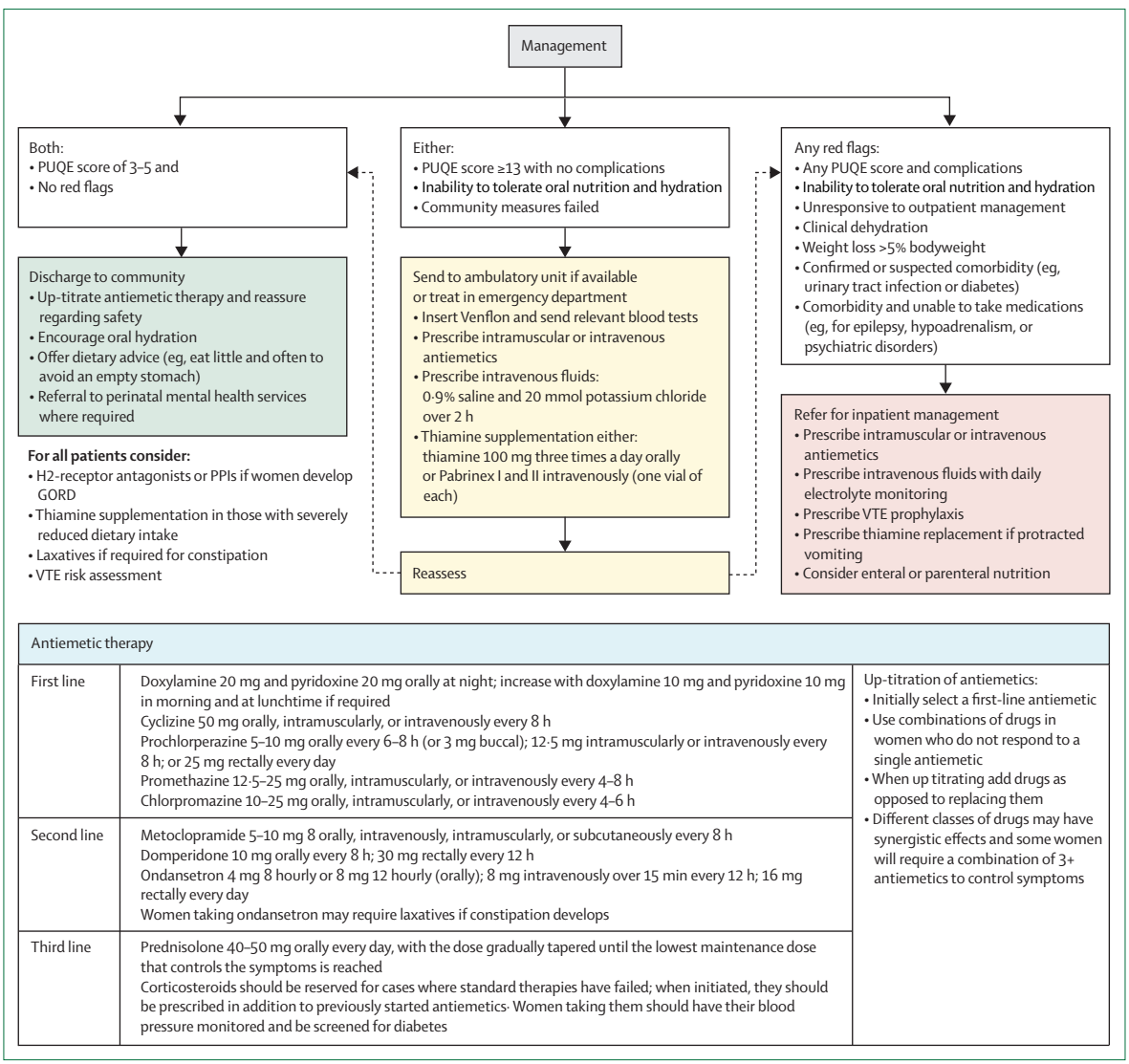

Figure 3: **Recommended management algorithm for hyperemesis gravidarum and nausea and vomiting of pregnancy**
GORD=gastro-oesophageal reflux disease. PPI=proton pump inhibitor. PUQE=pregnancy-unique quantification of emesis. VTE=venous thromboembolism. Adapted from the Royal College of Obstetricians and Gynaecologists' guideline.[63]

symptoms do not improve despite intravenous fluids and multiple, regular parenteral antiemetics. The rationale for corticosteroids being reserved for third-line treatment is that they are associated with an increased risk of developing steroid-induced gestational diabetes, hypertension, maternal infection, and preterm premature rupture of membranes.[74,75] Where used, women should therefore be screened for these conditions. If corticosteroids are beneficial, a gradual dose reduction over weeks or months is recommended until the minimum effective dose is reached to prevent recurrence of symptoms. Women taking at least 5 mg of prednisolone for more than 3 weeks should receive stress doses of hydrocortisone in labour as they are at risk of having developed adrenal insufficiency.[76]

Medications from different classes can work synergistically. If first-line drugs are ineffective then second-line drugs should be started; however, if first-line medications are partly effective, combination therapy with second-line medications is appropriate. Antiemetic therapy is required until the gestation at which symptoms would have abated spontaneously. Women with severe hyperemesis gravidarum will often require multiple antiemetics, sometimes until delivery. Attempts to reduce or discontinue therapy prematurely can result in distressing recurrence of symptoms.

The electrolyte disturbances described earlier mandate adequate sodium and potassium replacement, and 3 L per day of 0·9% saline is recommended, typically with 40 mmol of potassium chloride in each litre bag.

Daily monitoring of electrolytes is important to ensure adequate correction and to avoid complications of electrolyte disturbance. The use of dextrose-containing fluids is not appropriate as they do not contain adequate amounts of sodium, and high concentrations of dextrose before thiamine replacement can precipitate Wernicke's encephalopathy.[47]

To reduce the risk of Wernicke's encephalopathy, thiamine supplementation with either oral thiamine (100 mg three times daily, which is poorly tolerated) or Pabrinex I and II (one vial of each weekly until eating regularly) is recommended in women with protracted vomiting or in those unable to tolerate prenatal vitamins or foods rich in vitamin B1.

Symptoms of GORD can be treated with conservative management, including elevating the head of the woman's bed and avoiding irritant foods or eating late at night. Some women will require pharmacological therapies—such as antacids, H2-receptor antagonists, and proton pump inhibitors—that have been shown to reduce the severity of NVP.[77] All women with hyperemesis gravidarum should also be assessed for their risk of venous thromboembolism, according to UK-based guidance. No published data have so far shown an effective treatment for the management of ptyalism; clonidine hydrochloride at a dose of 0·1 mg daily as required has been successfully used in one case report.[78] Repeated vomiting can result in dental decay and women with hyperemesis gravidarum should therefore be advised to consult a dentist.

Dietitians should be consulted regarding nutritional support. Oral nutritional supplements can supplement an inadequate food intake or provide a more balanced nutritional intake for women who can only tolerate a small range of foods. Those who continue to be symptomatic into the late second or third trimester should also be offered serial fetal growth scans.

There are scarce data for the effectiveness of enteral and parenteral feeding in hyperemesis gravidarum. Enteral tube feeding options with nasogastric, nasoduodenal, or nasojejunal tubes, or percutaneous endoscopic gastrostomy or gastrojejunostomy feeding, should only be considered in consultation with gastroenterologists and with a multidisciplinary approach, once all classes of antiemetics and intravenous fluids have failed to alleviate symptoms. Parenteral feeding with a peripherally inserted central catheter is often better tolerated than enteral feeding; however, it carries a higher risk of infection and vascular complications. For the rare cases in which protracted parenteral nutrition is used, appropriate replacement of vitamins, essential fatty acids, and proteins is essential.

Information about culturally sensitive patient support groups should be provided to all women with NVP or hyperemesis gravidarum. A mental health assessment should also be included in the holistic management of women with the condition, and referrals should be made for psychological support where necessary. Pre-existing mental health conditions can be exacerbated by hyperemesis gravidarum, particularly if women are unable to keep down their medications (eg, antidepressants). Appropriate titration of antiemetic therapy is required to ensure women with comorbidity can tolerate their regular or time-critical medications and, in some cases, alternative routes of administration might need to be considered.

Termination of pregnancy should only be offered if the symptoms continue despite all measures having been exhausted. A termination request should not be taken to indicate that the pregnancy is not wanted or planned, but

***Panel:* Future directions and research priorities**

Priorities for research on hyperemesis gravidarum, as defined by patients and health-care professionals, were identified through a patient–clinician James Lind Alliance partnership and include:[96]

What causes hyperemesis gravidarum?
Further studies on the causal effect of GDF15 in different populations would be important, as well as exploring other genetic predisposing factors for hyperemesis gravidarum.

Can we find a cure?
Improved understanding of the aetiology of hyperemesis gravidarum will enable the development of cures. The discovery of the hormone GDF15 as a likely contributor to aetiology, and the potential to block or modify the response to it, offers exciting opportunities to develop new therapies for hyperemesis gravidarum. Novel therapies that target GFRAL and GDF15 signalling could cure hyperemesis gravidarum in women with a genetic susceptibility related to GDF15 signalling. Future investigation of other genes implicated in the condition could provide information to develop novel therapies.

How can we most effectively manage hyperemesis gravidarum?
To date, few studies have compared the efficacy of different antiemetic therapies or combinations thereof. A personalised approach to treatment is likely to be appropriate but has not yet been the subject of research studies. Very few studies have explored the efficacy of intravenous fluids in either the inpatient or outpatient setting. Further studies are required to establish whether nutrient replacement (informed by knowledge of deficiencies in individual women) could reduce the rate of adverse maternal and fetal outcomes. In addition to physical health complications, hyperemesis gravidarum has a notable, detrimental effect on maternal mental health; further studies should also establish strategies to manage this effect.

Is hyperemesis gravidarum preventable? What are the effects of preventive treatment or early intervention on the severity and duration of the condition in subsequent pregnancies?
To date, only one small study has explored the effect of pre-emptive antiemetics on the severity of hyperemesis gravidarum; larger studies are needed, including those with different antiemetics, to validate the study's findings of beneficial effects.

What are the immediate and long-term effects of hyperemesis gravidarum on the developing fetus?
Studies to date show heterogeneous outcomes with regard to immediate effects of the condition on the developing fetus. Further studies are required to confirm these findings and to determine underlying mechanisms for both the immediate and long-term effects of hyperemesis gravidarum to improve outcomes.

Future research should also explore ethnic disparities in care for women with hyperemesis gravidarum and the cultural perception of medications in pregnancy. This research will be valuable in ensuring all women with the condition can access equitable care.

rather as an indication of the desperation women with hyperemesis gravidarum feel when their symptoms are not validated and treated promptly and appropriately.

Women planning subsequent pregnancies have a high risk of recurring hyperemesis gravidarum. Family planning can be affected by the fear of recurrence: 40% of pregnancies can be postponed for this reason,[79] and some women limit their family size to avoid another pregnancy with hyperemesis gravidarum.[8] Pre-emptive use of antiemetics has been shown to reduce symptoms in further pregnancies and should therefore be considered.[80] The study by Koren and colleagues showed that, in women with a history of severe hyperemesis gravidarum, pre-emptive use of doxylamine and pyridoxine reduced the severity and duration of symptoms in a subsequent pregnancy.[80] Although this study is small and focused on a single antiemetic, the pre-emptive initiation of any antiemetic is likely to be beneficial.

The role of pre-pregnancy counselling in women with previous hyperemesis gravidarum is unclear, but gives clinicians the opportunity to discuss pre-emptive treatment and make a robust plan for management in subsequent pregnancies. Women with a previous history of hyperemesis gravidarum should not experience a delay in accessing antiemetics in future pregnancies. Considering the high recurrence risk, prescribing antiemetics in preparation for subsequent pregnancy could be beneficial.

Long-term effects of hyperemesis gravidarum

Maternal

Hyperemesis gravidarum can have long-term implications for maternal health. For women with the condition, up to 20% are left with symptoms of PTSS.[51,81,82] Worse symptoms of hyperemesis gravidarum are associated with higher PTSS symptom scores. Emotional distress,[83] depression, and anxiety are more common after pregnancies affected by hyperemesis gravidarum than after unaffected pregnancies.[82,84] Moreover, a small but noteworthy proportion of women affected by the condition are newly diagnosed with an eating disorder after pregnancy.[85] These findings, combined with the direct ill effects on maternal mental health during pregnancy, emphasise the need for adequate provision of continued mental health support for women recovering from a pregnancy affected by hyperemesis gravidarum.

The trajectory of physical recovery and long-term effects after a pregnancy affected by the condition has been scarcely studied. Two studies suggested increased incidence of colorectal, bladder, and thyroid cancers among women with a history of hyperemesis gravidarum.[86,87] Evidence for increased cardiovascular disease risk after a pregnancy affected by the condition is mixed.[88,89]

Offspring

Hyperemesis gravidarum is associated with severe maternal undernutrition,[34,44] which is known to have detrimental and long-lasting effects on offspring health. Findings among cohorts of men and women exposed to severe maternal undernutrition during the Dutch famine in 1944–45 were seminal in identifying the first trimester as the time of pregnancy where maternal undernutrition was associated with the broadest range of adverse outcomes. Such effects include increased cardiometabolic disease, respiratory disease, and cognitive decline, and a reduced lifespan.[90]

Long-term health effects in the offspring of women with hyperemesis gravidarum were summarised in a systematic review.[91] The authors identified a possible small increase in neurodevelopmental and mental health disorders, although there were concerns about the quality and size of the body of literature on which these conclusions were based.[91] The review also found a possible increase in testicular cancer among the offspring. More recently, a single study that was not included in the systematic review found a small increase in the risk of childhood cancer in the offspring of women with hyperemesis gravidarum.[92] Another study[93] found an increased chance of respiratory disease in childhood, although this effect could also be the result of confounding by sociodemographic factors.[94] A third large population-based study found that children affected by maternal hyperemesis gravidarum were more likely than children of unaffected mothers to require hospital admission in childhood for various reasons, including developmental, atopic, and digestive diseases.[95]

Conclusion

Hyperemesis gravidarum is a distressing, debilitating, and potentially fatal complication of pregnancy, and represents the most common reason for hospital admission in the first trimester. The condition is associated with several short-term and long-term adverse outcomes for the mother and her offspring. Women with hyperemesis gravidarum require careful monitoring of their physical and mental health to prevent its many potential complications. The condition is treated with intravenous fluid and electrolyte replacement and antiemetics, which can be used in combination and throughout pregnancy. Future directions and research priorities are summarised in the panel. Updated evidence-based guidelines exist, which, if followed by all health-care professionals, should improve the currently widespread experience of poor care by women with hyperemesis gravidarum.

Contributors

CN-P was invited to write the Review. All authors were involved in the literature search, agreed the structure of the paper, and contributed to the first draft of the manuscript. MN created the tables and figures. All reviewed the final manuscript before submission.

Declaration of interests

We declare no competing interests.

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